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Characterising Rainfall-Induced Soil Water Dynamics in Soil Profiles and Quantifying the Influencing Factors Over Mainland China

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ABSTRACT

Understanding rainfall-induced soil water dynamics (rainfall-induced SWD) is crucial for improving Earth System Models (ESMs), which have been limited in large-scale analyses. Using hourly data from 594 in situ stations, this study characterises the profile distribution patterns of rainfall-induced SWD, examines the environmental impacts and their scale effects and analyses the occurrence and controlling factors of preferential flow (PF). In the soil profile, three distinct patterns of soil water response to rainfall events were identified: a predominant shallow-decline pattern, a uniform pattern in barren land and a multi-phase pattern in areas with shallow groundwater, influenced by interactions between soil water and groundwater. The influence of rainfall and clay on soil water response amplitude decreases, while antecedent soil water content (SWC) and soil organic carbon (SOC) have increasing effects as the spatial scale decreases. Notably, these changes manifest as significant, stepwise shifts at critical scales, which should be considered in future investigations. The normalised difference vegetation index (NDVI) affects the maximum rate of the soil-wetting curve (Smax) in a bimodal fashion, with stronger impacts observed at regional scales and below $90 \times 90\,km^2$. Two distinct PF frequency distribution patterns emerge across soil layers: a consistent decrease with depth in PF frequency in the Changjiang river plain (CJ) region, corresponding with decreasing root density and microporosity. The predominant pattern is a mid-depth increase, caused by uneven grass root distribution, along with contributions from gravel content, cracks and soil fauna in deeper soils. Soil texture exhibits complex, nonlinear effects on PF frequency, varying with antecedent SWC in areas dominated by fine texture. Bulk density has a threshold effect: it negatively impacts PF frequency below $1.33\,g/cm^3$ and positively impacts it above $1.41\,g/cm^3$ in mainland China. Our results reveal novel profile patterns of rainfall-induced SWD and PF, providing fresh insights into the soil wetting process. This understanding is beneficial for effective model parameterisation and validation, ultimately enhancing the simulation of soil water dynamics in ESMs.

1 | Introduction

The critical zone is defined as the delicate layer of the Earth that extends from the top of the vegetation canopy to the bottom of the active groundwater zone (Brantley et al. 2007; Fan 2015; Han et al. 2021). It encompasses the biosphere, the entire pedosphere and portions of the hydrosphere, atmosphere and shallow lithosphere (Lin 2010; Wang et al. 2024; Zhang et al. 2025). Soil water content serves as a central medium that connects and regulates material and energy fluxes across the critical zone (Kumar et al. 2018; Wang et al. 2024). Acting as a primary regulator of critical zone functioning, soil water directly governs critical processes, such as the partitioning of water and energy between the land surface and atmosphere boundary, the solutes transport and the vitality of ecosystems (Fu et al. 2024; Kannenberg et al. 2024; Koster et al. 2004; Li, Li, et al. 2022; Singh et al. 2021). The soil water dynamics, encompassing the movement, storage and transformation of soil water, integrates hydrological, ecological and biogeochemical processes (Green and Erskine 2011; Kim and Lakshmi 2019; McColl et al. 2017), making them a key indicator of the critical zone's state and resilience. Consequently, advancing understanding of rainfall-induced soil water dynamics (rainfall-induced SWD) is crucial for improving Earth System Models (ESMs) (Hao and Chen 2024; Huang et al. 2016; Mohanty and Skaggs 2001; Singh et al. 2021; Tian et al. 2019; Vereecken et al. 2008).

The research investigating soil water dynamics can be classified into two categories: qualitative description and quantitative analysis. Qualitative descriptions involve the investigation of soil water dynamics through a qualitative comparison of the temporal variations of soil water values based on sparse in situ observations. These descriptions were typically conducted at the point scale (Bell et al. 2010; Farrick and Branfireun 2014; Qi et al. 2011), field or hillslope scale (Dong and Ochsner 2018; Fan et al. 2015; Green and Erskine 2011), or small catchment scale (Vinnikov et al. 1996). Quantitative analyses explore the spatiotemporal patterns of soil water dynamics through descriptive statistical methods, including probability distribution and standard deviation (Fan, Clark, et al. 2019; Tian et al. 2019), empirical orthogonal function (EOF) (Hu et al. 2017; Wang et al. 2017; Wang, Wu, and Zhan 2020), temporal stability analysis (TSA) (Wang, Wu, and Zhan 2020) and stored precipitation fraction (Kim and Lakshmi 2019; McColl et al. 2017). Quantitative analyses have been generally conducted on the small catchment scale (Hu et al. 2017; Hu et al. 2010), regional scale (Fan et al. 2015) and global scale (Kim and Lakshmi 2019; McColl et al. 2017). Most large-scale studies on soil water dynamics have been limited to surface layer due to constraints associated with aircraft-based remote sensing (Dong and Ochsner 2018; Li and Sawada 2022). Although these studies have provided foundational insights, the focus on shallow layers leaves significant gaps in understanding rainfall-induced SWD processes—knowledge that is crucial for enhancing land-atmosphere interactions within ESMs (Gao et al. 2020; Green and Erskine 2011; Li, Migliavacca, et al. 2022; Tian et al. 2019; Vereecken et al. 2022). To address this, continuous multi-depth in situ monitoring offers process-based quantification of rainfall-induced SWD, which can provide more detailed metrics for land-atmosphere feedbacks (Fan et al. 2015; Green and Erskine 2011; He et al. 2025; Singh et al. 2021; Tian et al. 2019; Xue et al. 2025).

To conduct a process-based analysis of rainfall-induced SWD in soil profiles, the concept of soil-wetting events was defined and has been applied to investigate rainfall-induced SWD with observations of 8 soil profiles in the Parapuños experimental catchment and 9 soil profiles in a small experimental watershed in the Iberian Peninsula (Lozano-Parra et al. 2015; Lozano-Parra et al. 2016), 8 soil profiles in the Tibet plateau in China (Gao et al. 2020), 32 soil profiles in the upper reaches of the Heihe river basin in northwest China (Tian et al. 2019) and three hillslopes in Nantahala national forest of western north Carolina, United States (Singh et al. 2021). All of these studies have focused on rainfall-induced SWD at small scales with limited in situ observations. However, rainfall-induced SWD synthesises the effects of land surface characteristics (soil properties, vegetations, topographic attributes, etc.) and atmospheric factors (rainfall) (Jarecke et al. 2021; Wang et al. 2017; Wang, Wu, and Zhan 2020; Yang et al. 2018), thus the variations in rainfall-induced SWD can be complex and may significantly vary from field to regional scales (Dong and Ochsner 2018; Fathololoumi et al. 2021; Kim and Lakshmi 2019). Therefore, it is crucial to characterise the profile distribution of rainfall-induced SWD at large scales.

Furthermore, despite the growing interest in studying the impacts of environmental factors on rainfall-induced SWD, there are still several main concerns that need to be addressed. First, the complex nature of environmental impacts and the significant scale effects due to cross-correlations among environmental factors necessitate a comprehensive investigation at various scales (Hu et al. 2017; Ruiz-Sinoga et al. 2011; Wu et al. 2024). Specifically, previous studies have primarily focused on the impacts of vegetation types (Gao et al. 2020; Li and Sawada 2022; Qiu et al. 2021; Tian et al. 2019) and topographic conditions (Lee and Kim 2022) at small scales. However, the impact of soil properties on rainfall-induced SWD has received little attention (Dong and Ochsner 2018; Jarecke et al. 2021; Qiu et al. 2013; Wang et al. 2017; Wang, Wu, and Zhan 2020), indicating a need for further investigation. Secondly, the occurrence and controlling factors of preferential flow (PF) in the context of rainfall-induced SWD are crucial for understanding hydrological processes and solute transport within the soil–plant–atmosphere continuum (Guo et al. 2018; Tang et al. 2020; Wiekenkamp et al. 2016; Zhang, Tu, et al. 2024). However, previous studies have primarily focused on PF at small scales, such as point-scale (Paradelo et al. 2016; Zhang, Tu, et al. 2024), hillslope-scale (Demand et al. 2019; Tang et al. 2020; Wiekenkamp et al. 2016) and catchment-scale (Kang et al. 2023). Consequently, these site-specific findings can conflict under varying conditions, thus leaving some local explanatory variables unidentified (Zhang, Tu, et al. 2024).

To summarise, the objectives of this study are: (1) to characterise the profile distribution of rainfall-induced SWD on a large scale, (2) to quantify the impacts of environmental factors on soil wetting processes and analyse their scale effects and (3) to analyse the occurrence and controlling factors of PF across mainland China. We collected hourly rainfall and soil water data from 594 in situ stations across mainland China from January 2017 to December 2019. Due to cross-correlation and data imbalance among environmental factors, it is a challenge to quantify the environmental impacts using only statistical methods. Therefore, machine learning methods were also applied to quantify these impacts in this study. The results would provide essential knowledge for rainfall-induced SWD at large scales and offer new insights into the development and improvement of ESMs.

2 | Materials and Methods

2.1 | Study Area

Mainland China is located between 73°–136° E and 18°–54° N, with a complex topography and strong variability of both the rainfall and soil water (Figure 1). Mainland China can be

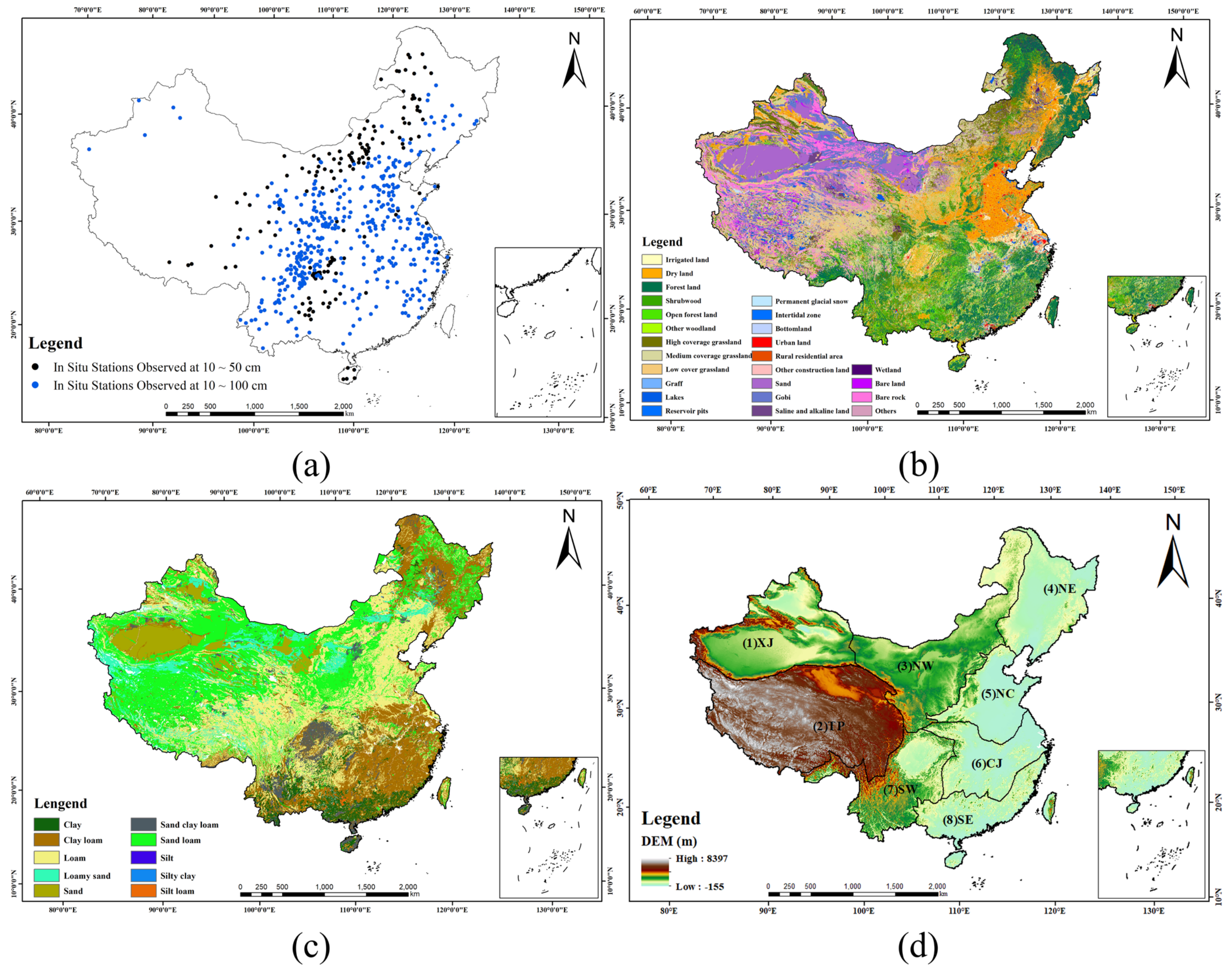

FIGURE 1 | Environmental information across mainland China. (a) In situ stations, (b) Land cover, (c) Soil texture, (d) Eight subregions. HCG refers to high-coverage grassland, MCG refers to medium-coverage grassland and LCG refers to low-coverage grassland.

divided into eight subregions depending on topography and climatic conditions, which were widely applied to comprehensively investigate hydrological processes in this area (Chao et al. 2021; Chen and Yuan 2020; Liu et al. 2022). The eight subregions are (1) Inland Xinjiang (XJ), (2) the Qinghai-Tibetan plateau (TP), (3) Northwestern China (NW), (4) Northeastern China (NE), (5) Northern China (NC), (6) the Changjiang river plain (CJ), (7) Southwestern China (SW) and (8) Southeastern China (SE) (Figure 1d). These eight subregions were utilised in this study.

2.2 | Data

We obtained hourly precipitation and soil water data at 594 in situ stations archived by the National Meteorological Information Center (NMIC) of the Chinese Meteorological Administration (CMA) from January 2017 to December 2019 across mainland China. Both rainfall and soil water stations are collocated. These data were carefully compiled by the NMIC of CMA (Figure 1), including extreme value examination, internal consistency checks and questionable data removal (Ling et al. 2021; Shen et al. 2010; Zhai et al. 2005). The soil water data were measured by DZN2 (GStar-I) sensor at soil depths of 0–10 cm, 10–20 cm, 20–30 cm, 30–40 cm, 40–50 cm, 50–60 cm, 60–80 cm and 80–100 cm, and they are subsequently referred to layers 1–8 in this study. Due to various soil depths at different in situ stations, soil water data were measured at Layers 1–5 or at Layers 1–8 (Figure 1a). Furthermore, in order to ensure that only rainfall events were analysed, precipitation data with air temperatures above 0°C were selected. Additionally, to avoid the effects of freezing and thawing processes, only soil-wetting events with soil temperatures above 0°C in the soil profile were analysed (Gao et al. 2020). Notably, all in situ stations are set up on flat grasslands or barren lands in accordance with the establishment standards for in situ stations by the NMIC of the CMA. The slopes for the in situ stations are relatively low, with a mean of 4.80 and a median of 2.39 (Figure A1, see Appendix).

The soil texture, bulk density and soil organic carbon (SOC) data are provided by the national Tibetan plateau data center (TPDC, https://data.tpdc.ac.cn/en/data/46ddd893-3b2b-4bb3-b9e6-b043f3c5c3a2) at five soil layers with a resolution of

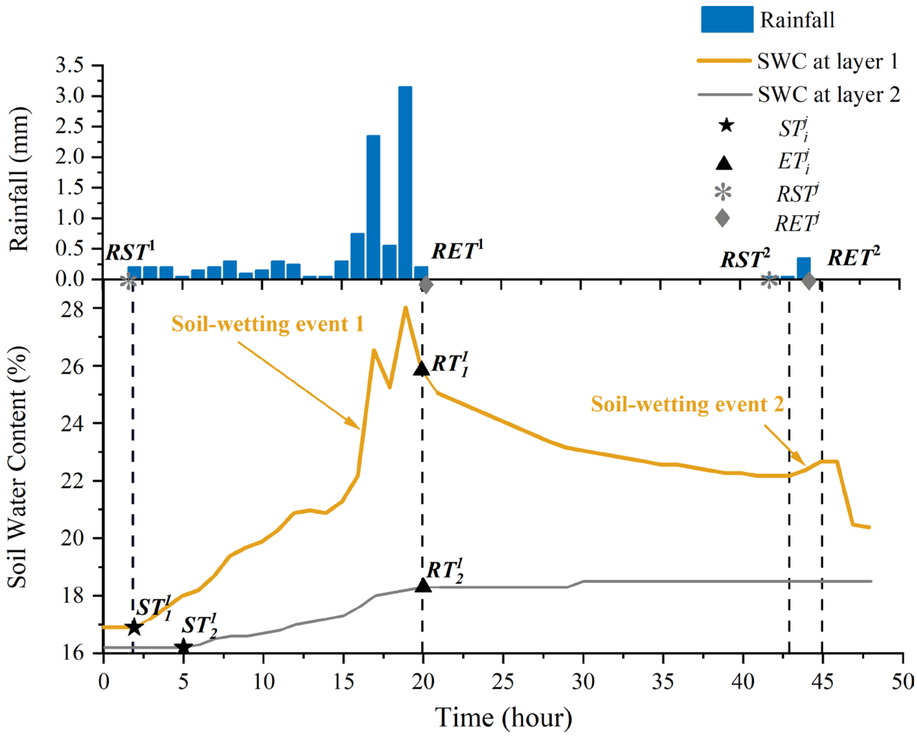

FIGURE 2 | Example of the identification of rainfall-induced soil water dynamics (SWD).

90 m × 90 m (Shi et al. 2025). The normalised difference vegetation index (NDVI) from moderate resolution imaging spectroradiometer (MODIS) MOD13Q1 product during 2017–2019 is provided by the Level-1 and Atmosphere Archive & Distribution System Distributed Active Archive Center (LAADS DAAC, https://ladsweb.modaps.eosdis.nasa.gov/) with a resolution of 250 m × 250 m. Both the soil properties and NDVI data in the grid where the in situ station is located were used as the representative data for that station.

2.3 | Identification of Rainfall-Induced SWD

In this study, we have classified the soil water responses to rainfall events into two categories (Figure 2): (1) Nonresponse events, which refer to events that rainfall does not result in any increase in soil water, and (2) Soil-wetting events, which refer to events that soil water experiences a significant increase (Lozano-Parra et al. 2015; McMillan and Srinivasan 2015). For each in situ station, we have quantitatively described the jth soil-wetting event at soil layer i based on the following four indices: accumulation of soil water increment of layer i ($ASWI_i^j$, vol%), the maximum rate of the soil wetting curve ($S\max_i^j$, vol%/h), the soil water lag time ($SWLT_i^j$, h) and the difference in SWLT between layer i and layer $i-1$ (DLT_i^j, h) (Gao et al. 2020; Tian et al. 2019). In this study, rainfall-induced SWD is analysed using these four indices related to soil-wetting events. These indices can be classified into two groups: response amplitude indices, including ASWI and Smax; and response time indices, including SWLT and DLT. The corresponding formulas are as follows.

$$ASWI_i^j = \sum_{t=ST_i^j}^{ET_i^j} \Delta\theta_i^t \tag{1}$$

$$S\max{}_i^j = \max\left(\frac{\Delta\theta_i^t}{\Delta t}\right) \tag{2}$$

$$SWLT_i^j = ST_i^j - RST^j \tag{3}$$

$$\mathrm{DLT}_i^j = \mathrm{SWLT}_i^j - \mathrm{SWLT}_{i-1}^j \tag{4}$$

with

$$\Delta\theta_i^t = \begin{cases} \theta_i^{t+\Delta t} - \theta_i^t, & \text{for } \theta_i^{t+\Delta t} - \theta_i^t > 0 \\ 0, & \text{for } \theta_i^{t+\Delta t} - \theta_i^t \le 0 \end{cases} \tag{5}$$

where $\theta_i^{t+\Delta t}$ and θ_i^t are soil water values at layer i at times $t + \Delta t$ and t, respectively, $ST_i^j \le t \le ET_i^j$ and Δt is 1 h in this study. As shown in Figure 2, for the jth soil-wetting event, RST^j is the time at which the jth rainfall event starts and RET^j is the time at which the jth rainfall event ends. ST_i^j is the start time of jth soil-wetting event at layer i, which is the time at which soil water first increases after RST^j; and ET_i^j is the end time of jth soil-wetting event at layer i, which is the time at which soil water first decreases or shows no change after RST^j. In this study, we have identified two soil-wetting events that were separated by a period ≥ 6 h with no effective increase of soil water between them (Lozano-Parra et al. 2015; Lozano-Parra et al. 2016; Tian et al. 2019). Moreover, continuous rainfall events were considered as a single rainfall event, and various rainfall events causing the same soil-wetting event were also considered as a single rainfall event. The total rainfall amount for a soil-wetting event is the cumulative sum of all contributing rainfall events. Only the increase in soil water after rainfall event starts was analysed to ensure that it is rainfall-induced SWD and exclude any soil water increase not caused by rainfall.

2.4 | Statistical Analysis of Rainfall-Induced SWD

Boxplots were used to describe the profile distribution of rainfall-induced SWD indices based on previous research (Gao et al. 2020; Lozano-Parra et al. 2016). One-way analysis of variance (ANOVA) ($p < 0.05$) was further used to test differences of rainfall-induced SWD indices between soil layers (Tian et al. 2019; Xue et al. 2025). When the differences in layers are significant, the least significant difference (LSD) post hoc-test method was subsequently applied to conduct multiple comparisons of means ($p < 0.05$) (Muenchen 2011; Krzywinski and Altman 2014; Tian et al. 2019). The empirical cumulative distribution function (CDF) was applied to describe data frequencies of rainfall characteristics across mainland China. The probability of soil water increasing in each soil layer is calculated as the proportion of occurrences where soil water increases in a particular soil layer when a soil-wetting event occurs in the profile.

2.5 | The Quantitative Analysis of Environmental Impacts

Based on their significant impacts and the attention they have received, the environmental factors investigated in this study include rainfall amount (mm), rainfall intensity (mm/h), rainfall duration (h), antecedent soil water (antecedent SWC, vol%), NDVI, the percentages of sand and clay, bulk density and SOC (Dong and Ochsner 2018; Gao et al. 2020; Guo et al. 2018; Jarecke et al. 2021; Tian et al. 2019). The atmospheric factors taken into account are rainfall amount, rainfall intensity and rainfall duration. The land surface characteristics considered are antecedent SWC, NDVI and soil properties. It is important to note that topographic effects were not considered in this analysis because all in situ stations were established on flat terrain.

2.5.1 | Analysis of the Impacts of Antecedent SWC

The antecedent SWC for a soil-wetting event caused by rainfall event starting at time *RST* is the soil water values at time $RST–\Delta t$ (Lozano-Parra et al. 2015). Typically, when simulating soil water dynamics at layer i, only the impacts of antecedent SWC at soil layers shallower than layer i or at layer i were considered in ESMs (Fan, Zhang, et al. 2019; Vereecken et al. 2022; Zhang, Tang, et al. 2024). However, the impact of PF is reported as a significant hydrological process that cannot be ignored (Guo et al. 2018; Lozano-Parra et al. 2016; Tang et al. 2020), thus it is necessary to analyse the impacts of antecedent SWC at all soil layers in the soil profile. In this study, at each station, Pearson correlation coefficients (r) were calculated between rainfall-induced SWD indices at layer i and the antecedent SWC at all soil layers in the soil profile, respectively. Moreover, for each rainfall-induced SWD index at layer i, the layer x with the highest correlation coefficient among all soil layers in the soil profile was defined as its largest impacting layer (LIL). The antecedent SWC at LIL at each station was applied to analyse the impacts of antecedent SWC in this study.

2.5.2 | Analysis of Environmental Impacts on the Spatial Patterns of Rainfall-Induced SWD and Their Scale Effects

Pearson correlation coefficients were calculated between the station-averaged indices and the environmental factors, to investigate the impacts of environmental factors on rainfall-induced SWD on a national scale, regional scales and various grid scales (Dong and Ochsner 2018; Peterson et al. 2019). The grid scales examined in this study range from $60 \times 60\,\text{km}^2$ to $1000 \times 1000\,\text{km}^2$. The selection of the scale range was based on both station numbers and grid numbers. At each scale, mainland China is divided into grids of the corresponding size. Within each grid, the correlation coefficients are calculated between station-averaged indices and environmental factors measured at stations located within that grid. Due to the varying number of stations in different grids, the correlation coefficients calculated cannot be directly compared. To ensure statistical robustness, Fisher's z-transformation was employed to derive Fisher's z-values based on correlation coefficients ($\text{z-value} = 0.5 \times \ln \frac{1+r}{1-r}$) (Pearson 1913; Zhu et al. 2023). The collective impact of environmental factors at this scale is then represented through boxplots of these derived Fisher's z-values. Finally, both ANOVA and the LSD methods ($p < 0.05$) were applied to analyse the differences in Fisher's z-values at different scales, aiming to assess the scale effects of each environmental factor.

2.6 | Identification and Analysis of Preferential Flow

Negative DLT values indicate a faster response in deeper soils compared to shallower soils, which can be attributed to PF (both vertical and lateral flow) or groundwater interaction (Fan, Zhang, et al. 2019; Hu et al. 2022; Wiekenkamp et al. 2016). When groundwater interaction occurs, there will be a soil water response in the bottom layer (Kang et al. 2023). Hence, the negative DLT values associated with soil water responses in the bottom layer were discarded in the following analysis. The frequency of PF events (F_{PF}) can be calculated by the proportion of PF events to total soil-wetting events in the soil profile, as indicated by formula (6). Furthermore, PF can be further divided into six types, denoted as $\text{PF}_{i\text{-}i+1}$ (i ranging from 1 to 6), according to the layer where layer $i+1$ responds faster than layer i. The frequency of $\text{PF}_{i\text{-}i+1}$ ($F_{PFi\text{-}i+1}$) is calculated by formula (7), which is applied to analyse the profile distribution pattern of PF occurrence.

$$F_{PF} = \frac{N_{PF}}{N_{soil-wetting}} \times 100\% \tag{6}$$

$$F_{PFi-i+1} = \frac{N_{PFi-i+1}}{N_{PF}} \times 100\% \tag{7}$$

where N_{PF} and $N_{soil\text{-}wetting}$ are the number of PF events and soil-wetting events in soil profile at each in situ station, respectively. $N_{PFi\text{-}i+1}$ is the number of $\text{PF}_{i\text{-}i+}$ events.

The spatial patterns of PF frequency were analysed based on F_{PF} across mainland China. Kernel density estimations (KDE) were employed to describe the frequency distributions of F_{PF} (Peterson et al. 2019). Regional differences of F_{PF} were assessed using ANOVA and the LSD method across eight subregions. Additionally, the profile distributions of PF frequency were analysed using ANOVA and the LSD method, based on $F_{PFi\text{-}i+1}$ for each soil layer across these subregions.

Spatial controls provide pathways for the PF events, temporal factors play a crucial role in determining the mechanisms that trigger PF occurrence (Guo et al. 2018; Kang et al. 2023). In other words, spatial factors determine where PF is more likely to happen, while temporal factors influence when it is actually initiated. In this study, correlation coefficients were calculated between F_{PF} and various environmental factors across mainland China and its subregions to investigate the spatial controls of PF events. These factors include station-averaged soil texture (the percentages of sand and clay), station-averaged bulk density, station-averaged SOC, station-averaged NDVI, station-averaged rainfall amount, station-averaged rainfall intensity and station-averaged antecedent SWC across eight soil layers. Regarding to temporal controls, since previous studies primarily focus on the conditions under which rainfall characteristics and antecedent SWC trigger the PF occurrence (Demand et al. 2019; Guo et al. 2018; Kang et al. 2023; Tang et al. 2020; Wiekenkamp et al. 2016; Zhang, Tu, et al. 2024) and few have investigated temporal factors on a large scale (Gao et al. 2018; Kang et al. 2023), the mechanisms across different regions remain a concern, with little consideration given to local variables such as vegetation and soil properties. To address these issues, the random forest (RF) method was applied to analyse the temporal controls of PF occurrence by identifying the relative importance of environmental factors across eight subregions. The RF method combines the ideas of decision trees and “bagging” ensemble learning (Breiman 2001), and excel at capturing the complex nonlinear relationship between environmental factors and PF occurrence (Kang et al. 2023; Singh et al. 2023). The relative importance is quantified by first building each tree using in-bag data, and then evaluating the importance of each variable by permuting it in the corresponding out-of-bag (OOB) samples and measuring the decrease in prediction accuracy. The environmental factors investigated include rainfall amount, rainfall intensity, NDVI, antecedent SWC, the percentages of sand and clay, bulk density and SOC at each soil layer. For output, a PF event in the soil profile was coded as 1, and its absence as 0. The RF analysis was performed using the sklearn (Scikit-learn) library in Python.

3 | Results

3.1 | Rainfall Characteristics During the Study Period

An analysis was conducted on all the rainfall events that caused soil-wetting events across mainland China from January 2017 to December 2019 (Figure 3). Spatially, the total rainfall amount for soil-wetting events during the study period was relatively high in the CJ, SW and SE regions, and low in the XJ, NW and NE regions (Figure 3a). However, the station-averaged rainfall amount and intensity causing soil-wetting events showed low values in the TP and XJ regions, while they were high in the other regions (Figure 3b,c). Furthermore, station-averaged rainfall amounts for soil-wetting events across mainland China ranged from 2.9 to 33.1 mm (Figure A2, see Appendix). Out of all the stations analysed, more than half recorded average rainfall amounts lower than 13.6 mm, while over 75% of the stations had average rainfall amounts lower than 16.9 mm (Figure A2). Station-averaged rainfall intensities for soil-wetting events ranged from 0.6 to 5.9 mm/h across all stations. For over half of the stations, the average rainfall intensities were found to be lower than 2.2 mm/h, while for more than 75% of the stations, they were lower than 2.6 mm/h (Figure A2). On a regional scale across the TP, NW, NE, NC, CJ, SW and SE regions, over half of the recorded average rainfall amounts were below 8.6, 11.4, 15.9, 16.4, 15.0, 11.7 and 18.6 mm (Figure A2). Additionally, more than 75% of stations reported average rainfall amounts lower than 10.0, 13.8, 16.9, 18.9, 17.8, 13.5 and 20.5 mm (Figure A2). Regarding average rainfall intensity, over half of the stations in the TP, NW, NE, NC, CJ, SW and SE regions recorded values below 1.4, 1.8, 2.6, 2.4, 2.3, 2.0 and 3.1 mm/h (Figure A2). Furthermore, over 75% of stations reported average intensities below 1.5, 2.1, 2.9, 3.0, 2.6, 2.3 and 3.8 mm/h (Figure A2).

3.2 | Profile Distributions of Rainfall-Induced SWD Across Mainland China

The numbers for soil-wetting events were high in south China and low in north China. While for more than half of the stations recorded soil-wetting event numbers lower than 110, 66, 47, 34, 24, 22, 16 and 11 at Layers 1–8, respectively. For more than 75% of stations, these numbers were lower than 199, 128, 98, 73, 59, 56, 46 and 36 at Layers 1–8. Additionally, as shown in Figure 4, the soil-wetting event numbers decreased with increasing soil depths and the probability of a soil water increase at each soil layer when there was a soil-wetting event in the soil profile also decreased with soil depth, consistent across both national and regional scales. The results suggest that soil-wetting events were more frequent in shallow soil layers, particularly in the arid region (TP, XJ and NW) (Figure 4b).

Figure 5 presents the boxplots of rainfall-induced SWD indices across specific soil depths in eight subregions, categorising them into three distinct profile distribution types of response amplitude. The uniform pattern is identified in the XJ region, where the indices ASWI and Smax remain consistent across soil depths. The shallow-decline pattern is observed in the SE, NE, SW, NC, NW and TP regions. In this pattern, notable declines in ASWI occur in Layers 1–7 (0–80 cm) in the SE region, Layers 1–6 (0–60 cm) in the NE region, Layers 1–5 (0–50 cm) in the SW, NC and NW regions and Layers 1–3 (0–30 cm) in the TP region ($p < 0.05$). Similarly, Smax demonstrates significant declines across layers 1–5 (0–50 cm) in the SE and SW regions and Layers 1–3 (0–30 cm) in the NC, TP, NW and NE regions ($p < 0.05$). Lastly, the multi-phase pattern is evident in the CJ region. This pattern shows a distinctive trend in ASWI, characterised by significant declines from Layers 1 to 3, stability from Layers 3 to 5, followed by a notable increase from Layers 6 to 8 ($p < 0.05$). Smax mirrors this trend, with significant declines

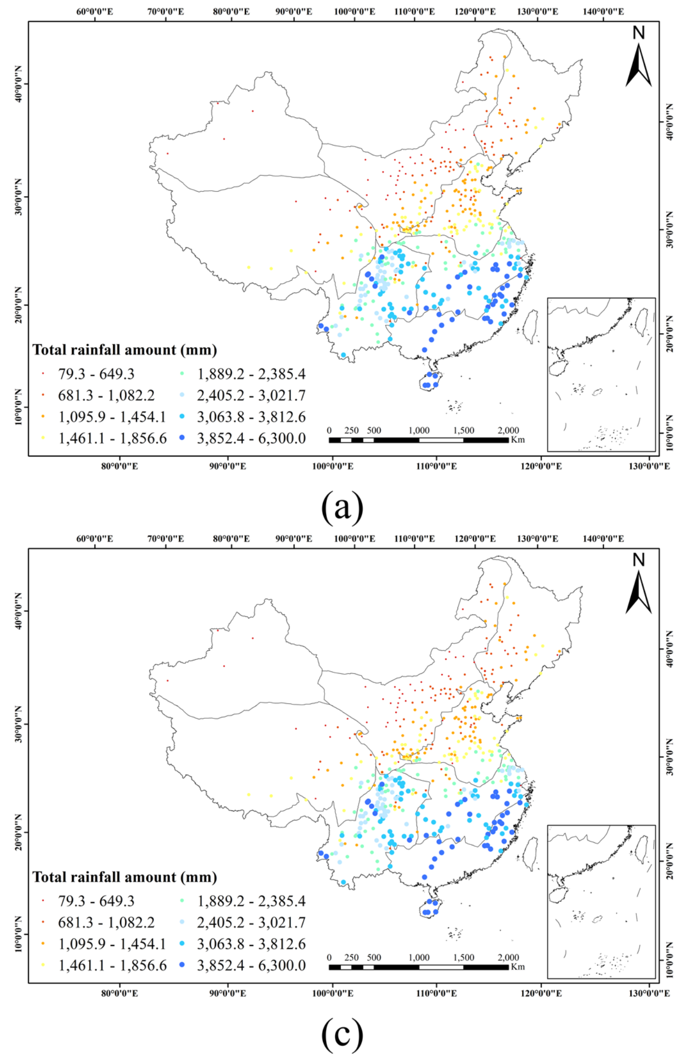

FIGURE 3 | Characteristics of rainfall events for soil-wetting events across mainland China. (a) Total rainfall amount during the study period, (b) Station-averaged rainfall amount, (c) Station-averaged rainfall intensity.

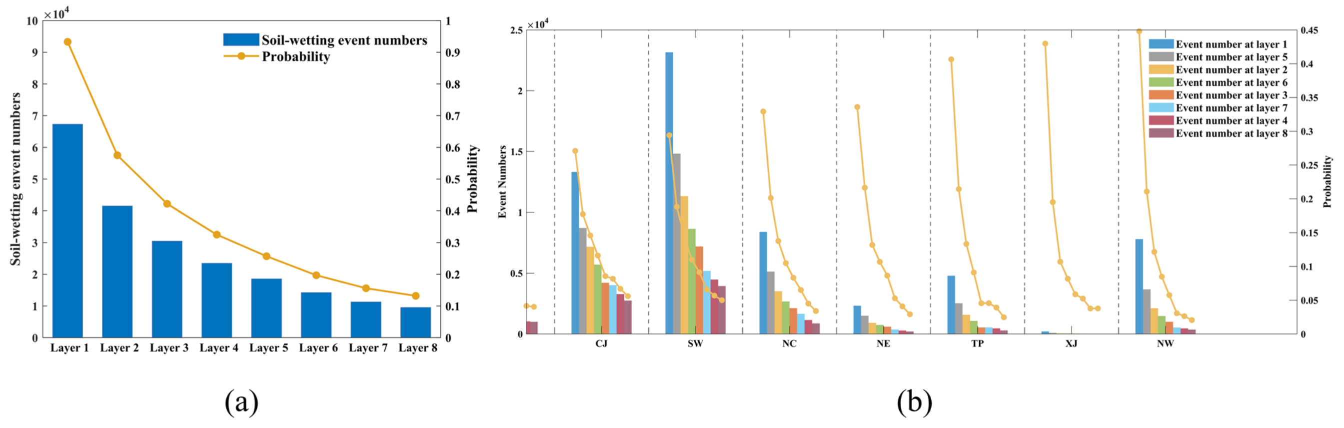

FIGURE 4 | Profile distribution of soil-wetting event numbers. (a) At the national scale, (b) At the regional scale. The probability of SWC increase in each soil layer is calculated as the ratio of occurrences where SWC increases in a particular soil layer when a soil-wetting event occurs in the profile.

from Layers 1 to 3, stable values from Layers 3 to 7, and a significant rise at Layer 8 ($p<0.05$). Although the ANOVA and LSD tests indicated that there are no significant differences in the mean values of the indices between some soil layers, the standard deviations exhibited distinct patterns (Figure A3, see Appendix). This reveals that even soil layers sharing a common average response can show significant differences in response stability and homogeneity.

FIGURE 5 | Profile distributions of rainfall-induced SWD in eight subregions. ASWI stands for accumulation of soil water increment and Smax stands for the maximum rate of soil water increase. The alphabet in the top refers to the group of indices at different soil layers tested by the ANOVA and LSD methods. Layers labelled with the same character indicate no statistically significant differences in rainfall-induced SWD indices between those soil layers, whereas layers with different characters exhibit significant differences. The letters 'a', 'b' and 'c' represent layers with decreasing levels of indices, arranged in the order of 'a' > 'b' > 'c'. Layers that share letters, such as 'ab' and 'bc', indicate intermediate states that are statistically transitional.

3.3 | Environmental Impacts on Spatial Patterns of Rainfall-Induced SWD and Their Scale Effects

On the national scale, rainfall is the most vital factor influencing rainfall-induced SWD indices (Figure 6a). Specifically, rainfall amount has the greatest effect on the ASWI; rainfall intensity exerts the most substantial impact on the Smax (Figure 6a). In addition, the response amplitude is significantly affected by antecedent SWC, NDVI and soil properties (sand, clay, bulk density and SOC) at a soil depth of 0–20 cm, with soil properties having the most substantial impact (Figure 6a). In contrast, at soil depths of 20–60 cm, only soil

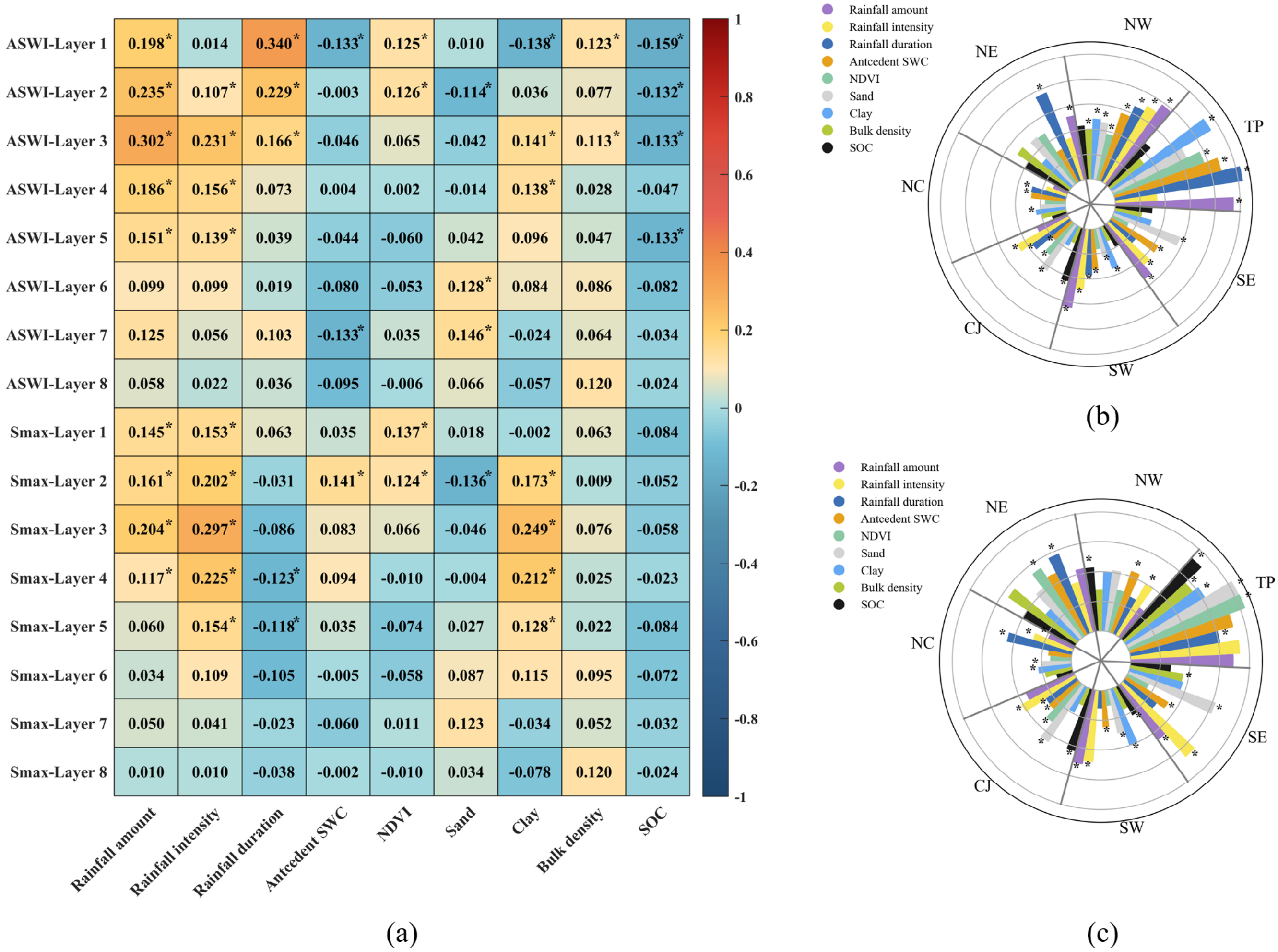

FIGURE 6 | Impacts of environmental factors. (a) On rainfall-induced SWD at the national scale, (b) On the accumulation of soil water increment (ASWI) at the regional scale, (c) On the maximum rate of the soil wetting curve (Smax) at the regional scale. * Refers to the r value passing the significance test with $p < 0.05$.

properties demonstrate significant effects (Figure 6a). On the regional scale, for ASWI, rainfall is the primary spatial factor, while the most important land surface characteristics include soil texture in the TP, CJ, SW and SE regions and antecedent SWC in the NW and NC regions (Figure 6b). Regarding Smax, rainfall is more critical than land surface characteristics in the NC, CJ, SW and SE regions (Figure 6c). Conversely, land surface characteristics are more significant in the TP, NW and NE regions (Figure 6c). The most important land surface characteristics include NDVI in the TP and NE regions, antecedent SWC in the NW region, whereas soil texture in the NC, SW and SE regions (Figure 6c).

As shown in Figure 7, rainfall amount has a tiered impact on ASWI, being strongest at the regional scale ('a'), weaker at 500×500–$600\times600\,\mathrm{km}^2$ ('b') and weakest at 60×60–$200\times200\,\mathrm{km}^2$ ('c'). For Smax, rainfall amount shows a critical threshold at $700\times700\,\mathrm{km}^2$, with uniformly high impacts ('a'/'ab') above this scale and a tiered weakening ('b') below it. Rainfall intensity consistently has moderate to high impacts on ASWI across all scales. However, its effect on Smax notably weakens from high ('a', 'ab') above $700\times700\,\mathrm{km}^2$ to moderate ('b') around $500\times500\,\mathrm{km}^2$ and low ('bc', 'c', 'abc') below $200\times200\,\mathrm{km}^2$. The impact of rainfall duration on ASWI decreases from high ('a') at the regional scale to moderate ('b') at 600×600–$1000\times1000\,\mathrm{km}^2$ and low ('c') below $300\times300\,\mathrm{km}^2$, while its effect on Smax remains invariant. The NDVI has scale-independent effects on ASWI but affects Smax in a bimodal pattern, intensifying at the regional scale and again below $90\times90\,\mathrm{km}^2$. Antecedent SWC impacts ASWI, increasing from low ('c') at the regional scale to high ('a', 'ab') below $100\times100\,\mathrm{km}^2$, with uniform effects on Smax. The influence of clay on ASWI varies without a clear trend, shifting from high ('a') for Smax at $900\times900\,\mathrm{km}^2$ to moderate ('b') at 400×400–$600\times600\,\mathrm{km}^2$ and low ('c') below $100\times100\,\mathrm{km}^2$. SOC effects on ASWI increase progressively from low ('c', 'bc') above $900\times900\,\mathrm{km}^2$ to moderate ('b') at 600×600–$700\times700\,\mathrm{km}^2$ and to strong ('a', 'ab') below $100\times100\,\mathrm{km}^2$, with uniform impacts on Smax.

In summary, as the scale decreases, the impacts of rainfall and clay on soil water response amplitude diminish, while antecedent SWC and SOC exert increasing influence. These changes are not smooth but occur in significant stepwise shifts at critical scales. The NDVI impacts Smax in a bimodal pattern, with greater effects observed at both regional scales and

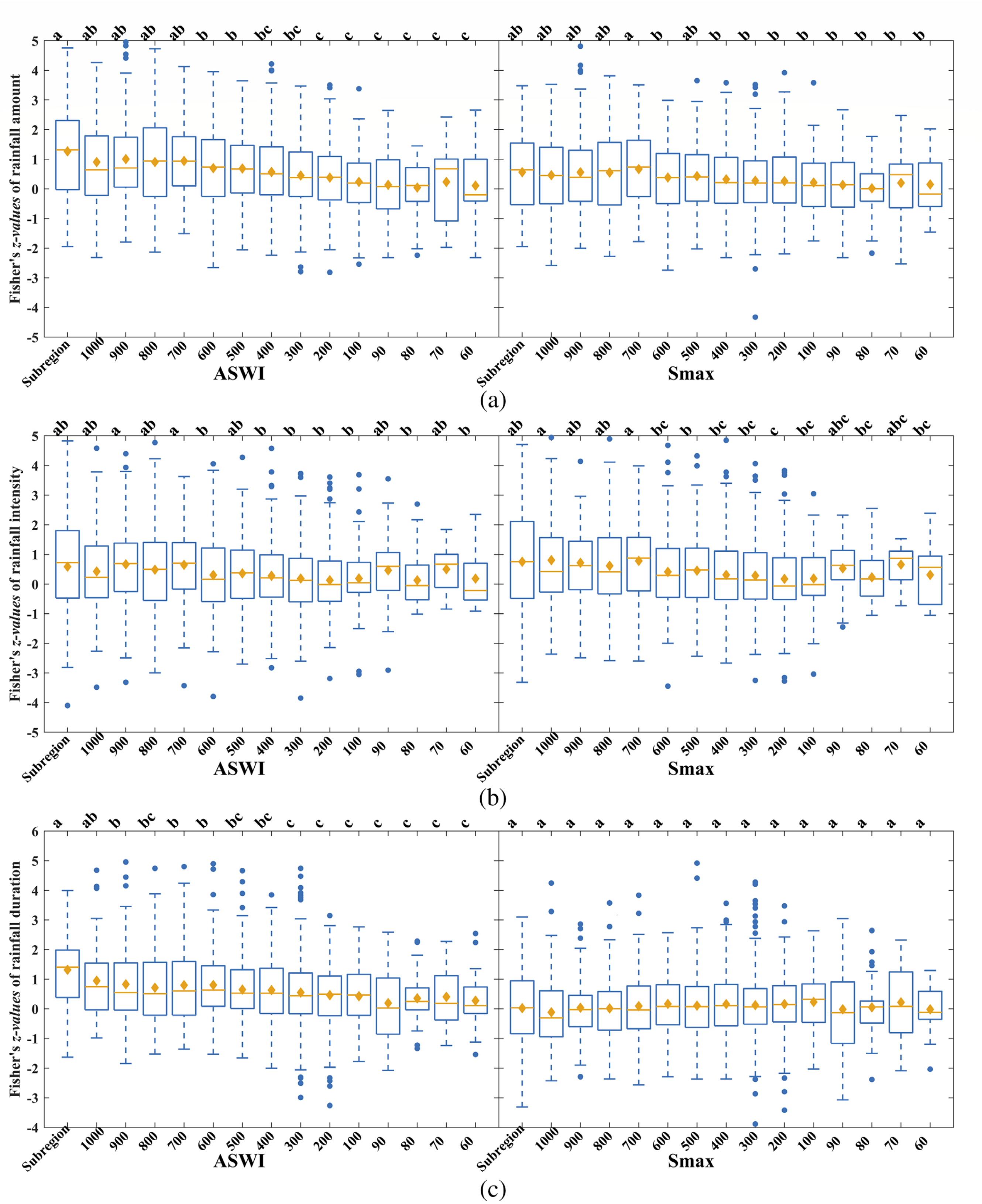

FIGURE 7 | Fisher's *z-values* of environmental factors on the spatial pattern of rainfall-induced SWD. (a) Rainfall amount, (b) Rainfall intensity, (c) Rainfall duration, (d) Normalised difference vegetation index (NDVI), (e) Antecedent soil water content (SWC), (f) Sand, (g) Clay, (h) Bulk density, (i) Soil organic carbon (SOC). The value of 1000 indicates a scale of 1000 km^2, and so on for subsequent values. The alphabet in the top refers to the group of Fisher's z-values at different scales tested by the ANOVA and LSD methods. Scales labelled with the same character indicate that there are no statistically significant differences in correlation coefficients between them, whereas scales labelled with different characters exhibit significant differences. The letters 'a', 'b' and 'c' represent groups with decreasing levels of environmental impact, arranged in the order of 'a' > 'b' > 'c'. Groups that share letters, such as 'ab' and 'bc', indicate intermediate states that are statistically transitional.

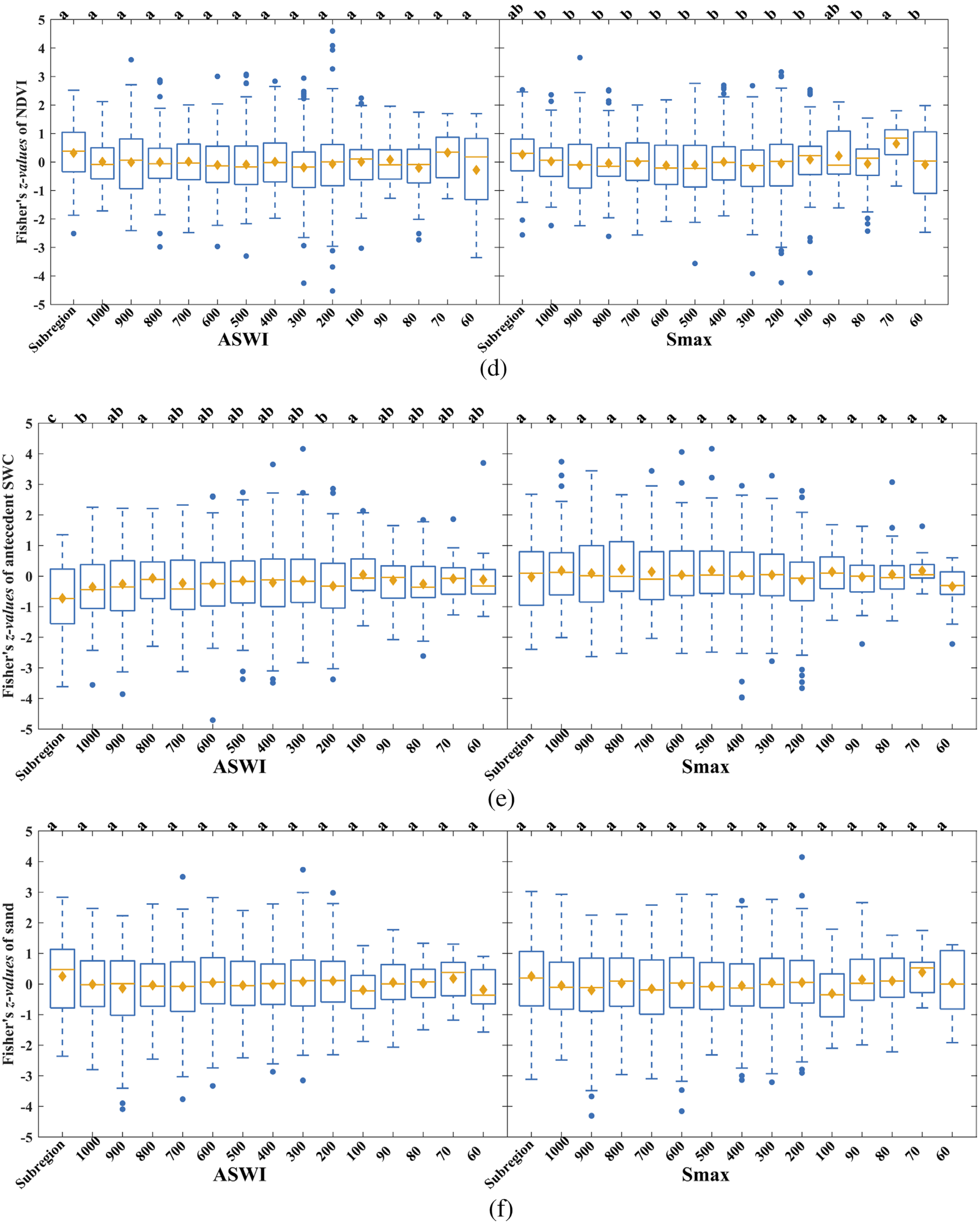

FIGURE 7 | (Continued)

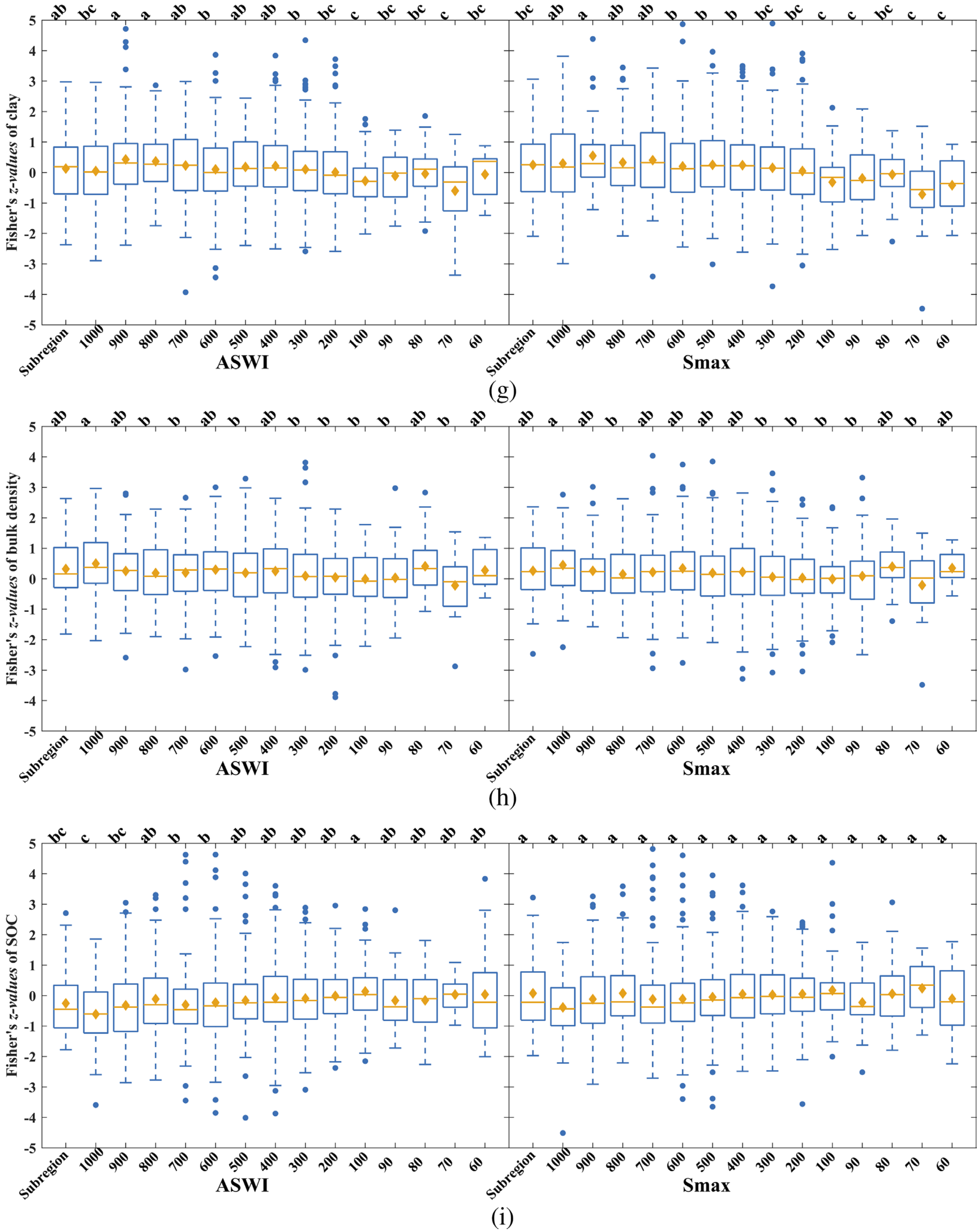

FIGURE 7 | (Continued)

below $90 \times 90\,km^2$. Additionally, sand and bulk density consistently affect soil water response amplitude, irrespective of the scale. These critical scales for the environmental impacts can be effectively implemented in ESMs by designing scale-aware parameterisation schemes that align with the spatial resolution of the model grid.

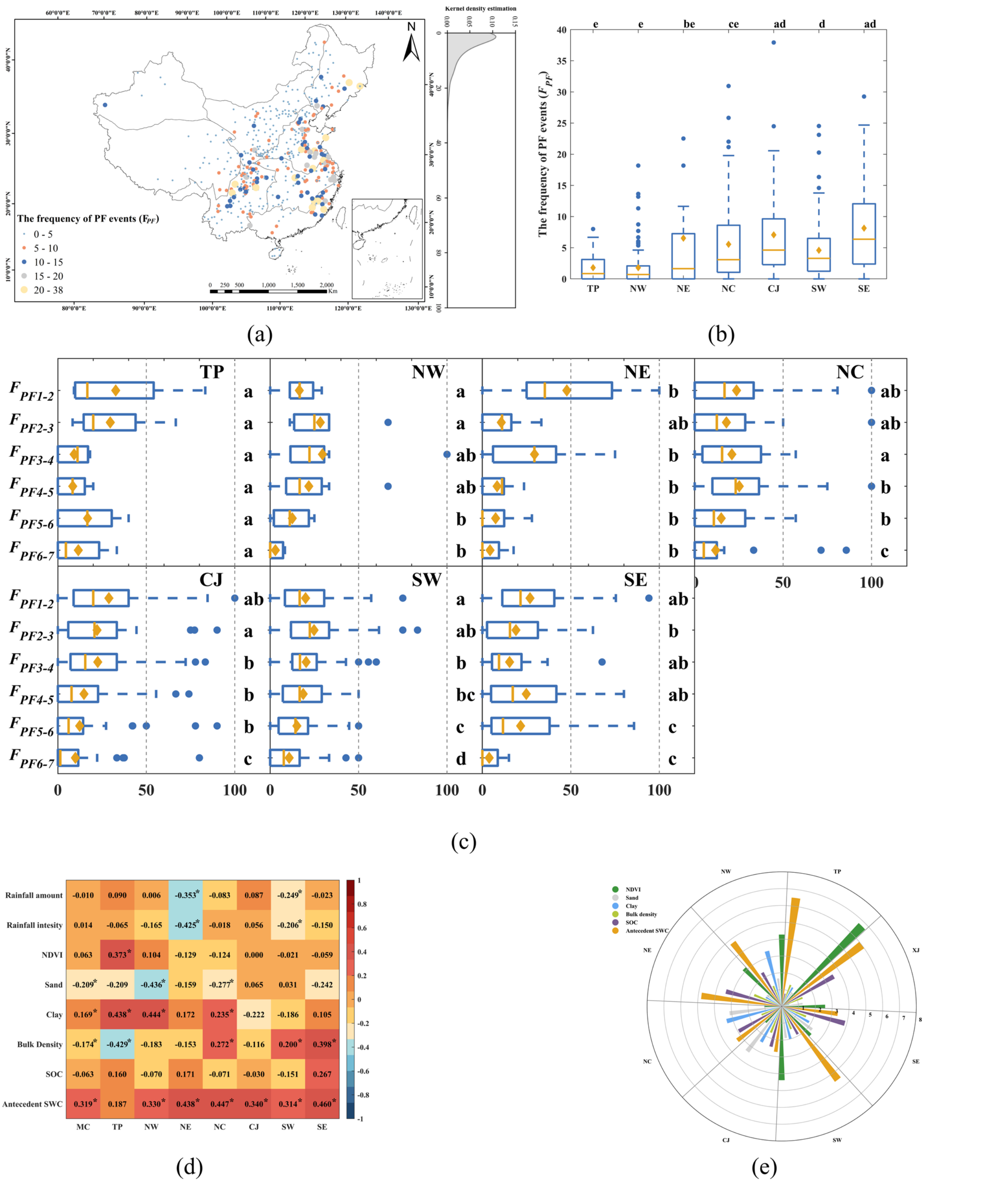

FIGURE 8 | The occurrence and controlling factors of preferential flow (PF) across mainland China. (a) The spatial patterns of PF frequency, (b) PF frequency on regional scale, (c) PF frequency at soil layers in eight subregions, (d) Spatial controls, (e) Temporal controls. *Refers to the *r*-value passing the significance test with $p < 0.05$.

3.4 | The Occurrence and Controlling Factors of Preferential Flow Across Mainland China

As shown in Figure 8a, the frequency of PF events (F_{PF}) ranges from 0% to 38% across mainland China, with the majority concentrated in the 0%–20% range. On the regional scale, the frequency of PF events is significantly higher in the CJ, SW and SE regions, while notably lower in the NC, NE, NW and TP regions ($p < 0.05$) (Figure 8a,b). Additionally, two distinct distribution patterns of PF frequency emerge across different soil layers (Figure 8c). The first pattern, observed in the CJ region, indicates a consistent decrease in PF frequency with increasing soil depth. The second pattern shows that PF frequency initially decreases with depth, then rises at a specific layer before continuing to decrease again. In this context, the specific layers are identified as follows: the 20–30 cm layer in the NW and

SW regions, the 30–40 cm layer in the NE region, the 40–50 cm layer in the NC and SE regions and the 50–60 cm layer in the TP region. It is important to note that the mid-depth increase observed in some regions did not reach statistical significance in the ANOVA and LSD post hoc tests, due to the outlier effects of PF frequency in those regions.

Spatial controls of the frequency of PF events were analysed by the correlation coefficients between station F_{PF} values and environmental factors (Figure 8d). Rainfall significantly influences PF frequency in the NE and SW regions; however, it does not have a notable effect on PF frequency across mainland China and other areas ($p < 0.05$). The NDVI positively affects PF frequency in the TP region ($p < 0.05$). Clay enhances PF frequency in the TP, NW and NC regions, as well as across mainland China ($p < 0.05$). In contrast, sand negatively influences PF frequency in the NW and NC regions, as well as across mainland China ($p < 0.05$). Bulk density exhibits a negative correlation with PF frequency in the TP region and throughout mainland China, while showing a positive correlation in the NC, SW and SE regions ($p < 0.05$). Additionally, SOC does not significantly affect PF frequency at either the national or regional scale ($p < 0.05$). Antecedent SWC consistently plays a crucial role in determining PF frequency on both national and regional scales, except in the TP region ($p < 0.05$). In summary, the most critical factors influencing PF frequency are soil texture in the TP and NW regions, while antecedent SWC affects mainland China and other regions ($p < 0.05$).

The temporal control triggering PF occurrence was examined by assessing the relative importance of environmental factors in eight subregions. It is evident that PF occurrence is primarily influenced by rainfall amounts on the regional scale (Figure A4, see Appendix). Additionally, as shown in Figure 8e, among the land surface characteristics, antecedent SWC, SOC, and NDVI exhibit obviously larger impacts on PF occurrence in the XJ, NE and SE regions. In the TP and SW regions, antecedent SWC and NDVI play crucial roles in PF occurrence. In the NW region, antecedent SWC, NDVI and soil texture (clay) demonstrate more pronounced impacts on PF occurrence compared to other factors. Moreover, in the NC and CJ regions, antecedent SWC, soil texture and SOC show more significant impacts on PF occurrence than other factors. In summary, the most critical land surface characteristics influencing PF occurrence include antecedent SWC in the XJ, TP, NW, NE, NC and SW regions, while soil texture is crucial in the CJ region, and SOC is significant in the SE region.

4 | Discussion

4.1 | The Profile Distribution Patterns and Controlling Mechanisms of Rainfall-Induced SWD Across Mainland China

There are three distinct profile distribution types of soil water response amplitude: the uniform pattern, the shallow-decline pattern and the multi-phase pattern. The uniform pattern is identified in the XJ region, can be attributed to the majority of in situ stations being located in barren land (Figure 1a,b), resulting in a consistent response across the soil profile. The shallow-decline pattern is characterised by a significant decrease in response amplitude with depth in specific shallow layers, while maintaining consistency in deeper soils. This pattern has been observed in previous studies conducted in the TP region (Dai et al. 2019; Tian et al. 2019). Furthermore, the most important land surface characteristics influencing soil water response in regions exhibiting this pattern include antecedent SWC, soil texture and NDVI (Figure 6b). Antecedent SWC defines the initial soil water deficit, thereby regulating the infiltration capacity, while soil texture controls the porosity and hydraulic conductivity that in turn determine the water retention capacity (Dingman 2015). The NDVI reflects vegetation effects on soil water dynamics through rainfall interception and root distribution (Filipović et al. 2020; Kang et al. 2023; Li et al. 2023). Previous research highlights that root distribution is a crucial factor regulating soil water response to rainfall (Tian et al. 2019; Kang et al. 2023; Zhang et al. 2025). In shallow soils, dense grass roots can reduce soil permeability (Li et al. 2023; Singh et al. 2018) while enhance water-holding capacity (Li et al. 2021; Tian et al. 2019; Zhang et al. 2025), resulting in a greater increase in soil water following rainfall (Dai et al. 2019; Tian et al. 2019; Zhang et al. 2025). As soil depth increases, root density typically decreases, leading to a reduction in water-holding capacity (Kang et al. 2023; Wang, Dong, et al. 2020). Therefore, this pattern is jointly influenced by antecedent SWC, soil texture and grass root distribution. It should be noted that this study does not quantitatively evaluate the impacts of root distribution, as accurate measurements across the soil profiles were not obtained. This topic warrants further investigation.

The multi-phase pattern, observed in the CJ region, also displays substantial declines in shallow soils. However, the response amplitude is significantly greater in the bottom layer (80–100 cm) compared to the intermediate layers (10–80 cm). To further investigate the multi-phase pattern, the rainfall-induced SWD is categorised into three distinct types: Type 1 exhibits responses in the upper and intermediate soil layers with no response at the bottom layer (85.4%); Type 2 shows responses exclusively in the bottom layer without intermediate layer involvement (4.8%); and Type 3 displays responses throughout the entire soil profile (9.8%). The CJ region is characterised by shallow groundwater tables, typically averaging less than 1 m in depth in the Yangtze River Estuary (PWRB 2015; Wang et al. 2024; Xie and Yang 2013). Mechanistic investigations suggest that Type 1 patterns occur under conditions of limited rainfall where the groundwater table remains hydrologically disconnected from the monitored soil layers (Wang et al. 2024). Under these circumstances, soil water dynamics are predominantly controlled by rainfall infiltration through soil matrix processes (He et al. 2022; Wang et al. 2024; Wittenberg et al. 2019; Xie and Yang 2013). The minimal topographic relief at the monitoring stations (Figure A1) indicates negligible contributions from lateral flow in this study, thus Type 2 patterns suggest direct groundwater recharge to deeper soil layers. This pattern occurs when the rising groundwater table enters the capillary fringe zone (generally within 60 cm below the monitored layers), enabling effective soil water replenishment through capillary action (Wang et al. 2024). Type 3 patterns represent comprehensive soil profile wetting, which develops under heavy rainfall conditions that promote both deep infiltration and groundwater table rise (Wang et al. 2024). In this

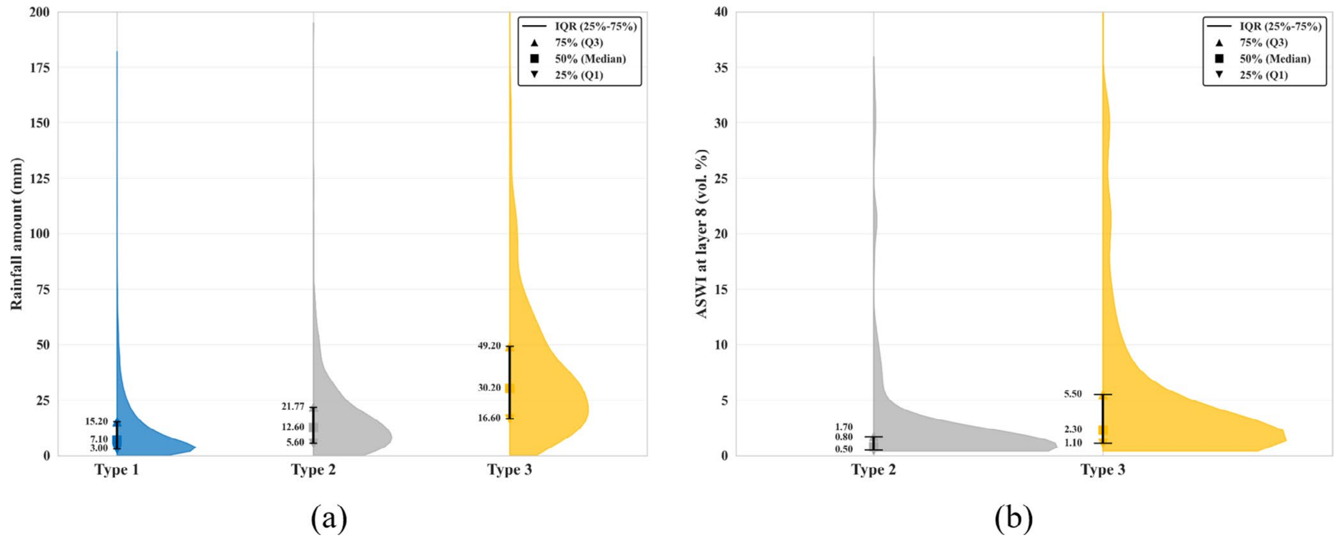

FIGURE 9 | Hydrological conditions for SWC response patterns in the CJ region. (a) Half violin plot of rainfall amount for three SWC response types, (b) ASWI at Layer 8 for Types 2 and 3.

study, rainfall analysis confirms that Type 1 is associated with the lowest rainfall amounts, Type 2 with moderate rainfall and Type 3 with the heaviest rainfall events (Figure 9a), providing partial validation of the proposed mechanisms. Furthermore, the significantly larger response amplitudes observed in Type 3 compared to Type 2 offer additional evidence supporting the combined recharge mechanism (Figure 9b). This study substantiates at the regional scale the soil water-groundwater interaction mechanisms identified by Wang et al. (2024) at the hillslope scale. Although Types 2 and 3 represent a small proportion of cases, they account for the most substantial response amplitudes in the bottom layer, highlighting groundwater's significant role in rainfall-induced SWD in the CJ region.

4.2 | Mechanisms for Controlling Factors Influencing Preferential Flow

To date, the variation of PF with soil depth at large scales remains unclear, limiting our understanding of the mechanism behind the relative macro-contribution of environmental factors to PF (Kang et al. 2023; Zhang, Tu, et al. 2024). Two distinct distribution patterns of PF frequency emerge across different soil layers. The first pattern, observed exclusively in the CJ region, indicates a consistent decrease in PF frequency with increasing soil depth. This trend correlates with a general decline in root density and macroporosity, which reduces the potential for PF occurrence (Kang et al. 2023; Wang, Dong, et al. 2020). Conversely, the second pattern features a mid-depth increase, which is the predominant pattern across mainland China. This finding confirms and greatly extends the local observation of Kang et al. (2022), who reported the significance of the 50–60 cm layer in the TP region. This study demonstrates that this mid-depth increase is a widespread phenomenon, consistently occurring across diverse regions including TP, NC, NE, NW, SW and SE. This pattern can be partially attributed to the uneven distribution of grassroots rather than a gradual decline. Dense grass root systems in the upper soil layers enhance the water-holding capacity (Tian et al. 2019; Zhang, Tang, et al. 2024; Zhang, Tu, et al. 2024), resulting in reduced PF frequency. In deeper soils, sparse grass roots play a crucial role in enhancing vertical flow by improving soil drainage and porosity, thereby creating conditions that facilitate vertical PF (Beven and Germann 1982; Li et al. 2023; Liu and Lin 2015; Wiekenkamp et al. 2016; Zhang et al. 2021). Additionally, the gravel content in deep soils in the TP region (Hu et al. 2022; Kang et al. 2022) and cracks created by rock fragments due to the karst topography in the SW region can enhance PF pathways (Chen et al. 2021). Furthermore, soil fauna, such as termite holes and earthworm burrows, also contribute to the increase in PF frequency (Demand et al. 2019; Zhang, Tu, et al. 2024). Collectively, these factors facilitate a more rapid response at specific soil layers. However, future studies will explore the quantitative analysis of their impacts.

Spatial controls provide pathways for the PF events, which means they determine where PF is more likely to happen (Guo et al. 2018; Kang et al. 2023). The NDVI positively influences PF frequency in the TP region. This is because higher vegetation coverage is indicative of a well-developed root system, which increases the abundance of soil pores, thereby improving soil permeability and PF frequency (Beven and Germann 1982; Jačka et al. 2021; Liu and Lin 2015; Wang et al. 2022; Wiekenkamp et al. 2016; Zhang et al. 2021). Soil texture significantly affects PF frequency in the TP, NW and NC regions but not in the NE, CJ, SW and SE regions. Soil textures can be categorised into three types: coarse texture, which includes sand and loamy sand; fine texture, which consists of clay, silty clay, clay loam or silty clay loam; and medium texture, which encompasses all other types (Jarvis et al. 2009). Significant impacts were detected in regions with uniform fine texture (23.0%), medium texture in the upper soil accompanied by fine texture in deeper soil (13.7%) and a combination of medium-fine and fine-medium textures (25.7%) in soil profiles. However, no significant impact was observed in areas dominated by uniform medium textures in soil profiles, which comprise 77.6% of those regions. This is further supported by findings of non-significant impacts in regions where medium and coarse textures predominate (Kang et al. 2022; Kang et al. 2023; Rahbeh 2019). Therefore, while previous studies showed inconsistent results regarding soil texture's influence on PF frequency (Kang et al. 2022; Kang et al. 2023; Rahbeh 2019), this study revealed that soil texture significantly influences PF frequency only in areas where fine textures are predominant. Notably, clay positively affects PF frequency in the TP, NW and NC regions. Positive effects occur in unsaturated soils (Demand et al. 2019; Grant et al. 2019; Zhang, Tu, et al. 2024) due to depositional clay layers that enhance soil aggregate stability and pore development in dry soils (Demand et al. 2019; Hudek et al. 2017), facilitating PF pathway formation (Chen et al. 2021; Kang et al. 2023). Conversely, clay negatively affects PF frequency in wet soils (Chen et al. 2021) because wet clay soils typically have lower permeability due to compact clay particles and high water retention capacity (Mosugu and Bradley 1999). This is proved by the non-significantly negative impacts in the CJ and SW regions in this study (Figure 8d). Therefore, soil texture exhibits complex, nonlinear effects on PF frequency, varying with antecedent SWC conditions in regions dominated by fine textures.

Bulk density is negatively correlated with PF frequency in the TP region, while it shows a positive correlation in the NC, SW and SE regions. This is because of the threshold effect for bulk density on PF frequency, as noted by Kang et al. (2022). Below this threshold, lower bulk density correlates with a higher number of macropores (Kang et al. 2022; Rahbeh 2019), resulting in a higher PF frequency. Conversely, above the threshold, higher bulk density is associated with a denser matrix, which can enhance PF occurrence (Katuwal et al. 2015; Nimmo 2021; Paradelo et al. 2016). In this study, negative impacts of bulk density are associated with an average bulk density value of $1.32 \pm 0.07\,g/cm^3$ in the TP region, while positive impacts are linked to average values of $1.38 \pm 0.02\,g/cm^3$, $1.37 \pm 0.05\,g/cm^3$ and $1.33 \pm 0.04\,g/cm^3$ in the NC, SW and SE regions, respectively. This finding is consistent with the positive effects observed at average bulk density values of 1.55 and $1.47\,g/cm^3$ (Katuwal et al. 2015; Koestel et al. 2013), while negative effects are seen at average values ranging from 0.98 to $1.41\,g/cm^3$ (Kang et al. 2022; Koestel et al. 2012). Therefore, based on the threshold effect proposed by Kang et al. (2022), we further identify these thresholds as being between 1.33 and $1.41\,g/cm^3$ (Figure A5, see Appendix). This threshold can be applied to classify soil columns within model grids, or to constrain pedotransfer functions that derive hydraulic properties from bulk density. Its explicit integration would improve the representation of soil water dynamics in ESMs, thereby enhancing the simulation of coupled hydrological processes.

The relationship between SOC and PF frequency is complex and strongly influenced by clay content, as these two components jointly govern soil structure (Dexter et al. 2008; Larsbo et al. 2016; Rahbeh 2019). At low SOC and clay levels (0.5–2 g/kg SOC and 10%–20% clay), SOC is positively correlated with PF, primarily due to macropore formation when the SOC/clay ratio approaches or exceeds 0.1 (Dexter et al. 2008; Rahbeh 2019). In contrast, at higher SOC levels, an increase in SOC reduces the number of macropores by promoting the development of micropores, which facilitate water infiltration and consequently inhibit PF occurrence (Larsbo et al. 2016; Rahbeh 2019). This dual effect helps explain the observed regional variability: negative impacts of SOC are evident in regions with either high SOC levels (CJ, SW) or low SOC levels combined with low SOC/clay ratios (NW, NC). In contrast, positive impacts occur in regions where the SOC/clay ratios are approaching or exceed 0.1 (TP, NE). Notably, the SE region stands out as an exception, displaying a positive effect despite its high SOC levels and low SOC/clay ratios. This underscores the fact that, at larger spatial scales, the intricate interplay between SOC and clay content often reduces the significance of SOC as a standalone factor in controlling PF dynamics.

4.3 | Implications and Limitations of This Study

In this study, all in situ stations are set up on flat grasslands or barren lands in accordance with the establishment standards for in situ stations by the NMIC of the CMA. As a result, the vegetation types and topographic attributes are similar, thereby minimising the influence of vegetation and topography in the analysis. However, these similar vegetation types and topographic attributes make it suitable for analysing the impacts of soil texture. Furthermore, considering that grasslands cover nearly 30% of the Earth's land surface area (Berdugo et al. 2020) and 40% of China's territory (Lyu et al. 2021), investigating rainfall-induced SWD on grasslands on large scales is essential for understanding global and regional water cycles.

In the analysis of environmental impacts, there is a scale mismatch between the soil properties data and the NDVI data at the grid scale, as well as with in situ observations at the point scale. Given the difficulty of obtaining in situ measurements of soil properties and NDVI at each in situ station, this study utilised the highest-resolution gridded datasets available (90 m for soil and 250 m for NDVI). By extracting pixel values at the precise locations of each in situ station, these high-resolution grids are effectively employed as ‘proxies’ for point-scale data. This approach significantly mitigates potential bias from scale mismatch compared to using coarser data. Furthermore, the analysis conducted across multiple grids at various scales demonstrates representativeness and generalisability, thereby ensuring the robustness of the analysis of environmental impacts in this study (Luo et al. 2017; Zhang et al. 2023).

Furthermore, this study qualitatively discussed the impacts of root distribution on rainfall-induced SWD and PF, along with the interaction mechanisms between soil water and groundwater based on findings from previous studies. However, quantitative assessments have been constrained by the limited availability of large-scale root distribution and groundwater level data. Future research will address this limitation through systematic monitoring and analysis.

5 | Conclusions

Using hourly rainfall and SWC data from 594 in situ stations across mainland China from January 2017 to December 2019, this study analysed the profile distribution patterns of rainfall-induced SWD, examined the environmental impacts and their scale effects and investigated the occurrence and controlling factors of PF. The main findings are as follows:

1. During rainfall events, three distinct patterns of soil water response amplitude were identified: a predominant shallow-decline pattern, a uniform pattern in barren land and a multi-phase pattern in areas with shallow groundwater, influenced by interactions between soil water and groundwater.
2. The influence of rainfall and clay on soil water response amplitude decreases, while antecedent SWC and SOC have increasing effects as the spatial scale decreases. Notably, these changes occur in significant, stepwise shifts at critical scales. The NDVI affects Smax in a bimodal fashion, with stronger impacts observed at the regional scale and below $90 \times 90\,\mathrm{km}^2$. Additionally, sand content and bulk density consistently influence soil water response amplitude across all scales.
3. Two distinct PF frequency distribution patterns emerge across soil layers: a consistent decrease in PF frequency in the CJ region, corresponding with decreasing root density and microporosity. The predominant pattern is the mid-depth increase, caused by uneven grass root distribution, as well as contributions from gravel content, cracks and soil fauna in deeper soils.
4. Spatial controls on PF events reveal that soil texture has complex, nonlinear effects on PF frequency, varying with antecedent SWC in areas dominated by fine texture. Bulk density has a threshold effect: it negatively impacts PF frequency below $1.33\,\mathrm{g/cm}^3$ and positively impacts it above $1.41\,\mathrm{g/cm}^3$ in mainland China. For better predictions of PF occurrence, soil texture and SOC should be incorporated into ESMs, alongside rainfall characteristics and antecedent SWC.

This comprehensive investigation offers valuable insights into rainfall-induced SWD at large scales, crucial for advancing research in ecohydrology and land surface-atmosphere interactions. The findings, including the critical scales of environmental impacts, the thresholds of bulk density and the temporal controls triggering PF occurrence, help for effective model parameterisation and validation for enhancing the simulation of soil water dynamics in ESMs.

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are available on request from the corresponding author. The data are not publicly available due to privacy or ethical restrictions.

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Appendix A

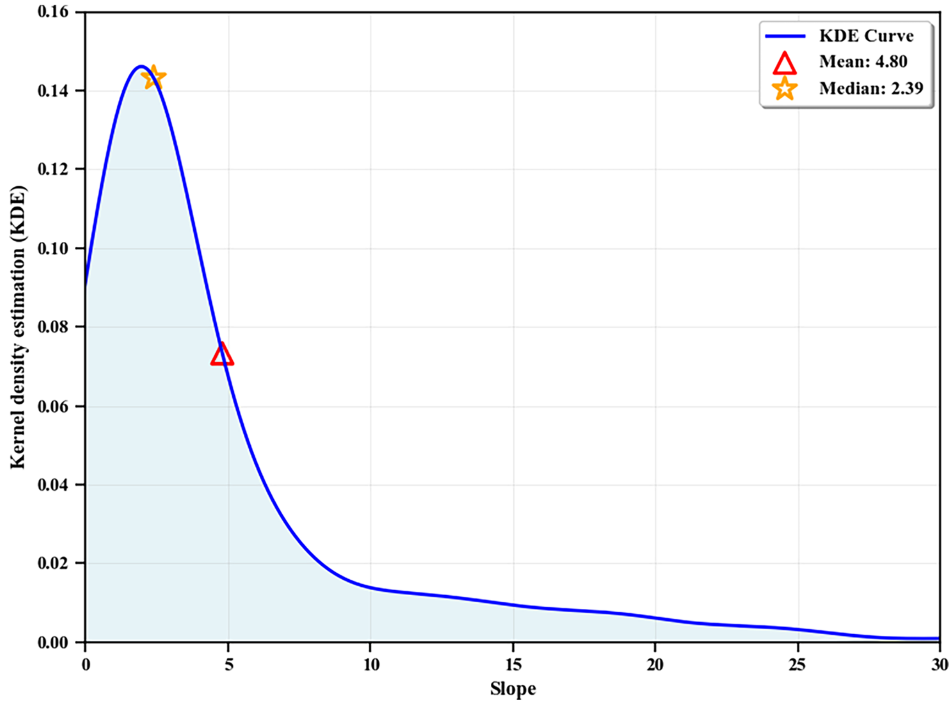

FIGURE A1 | Kernel density estimation (KDE) of slopes at in situ stations.

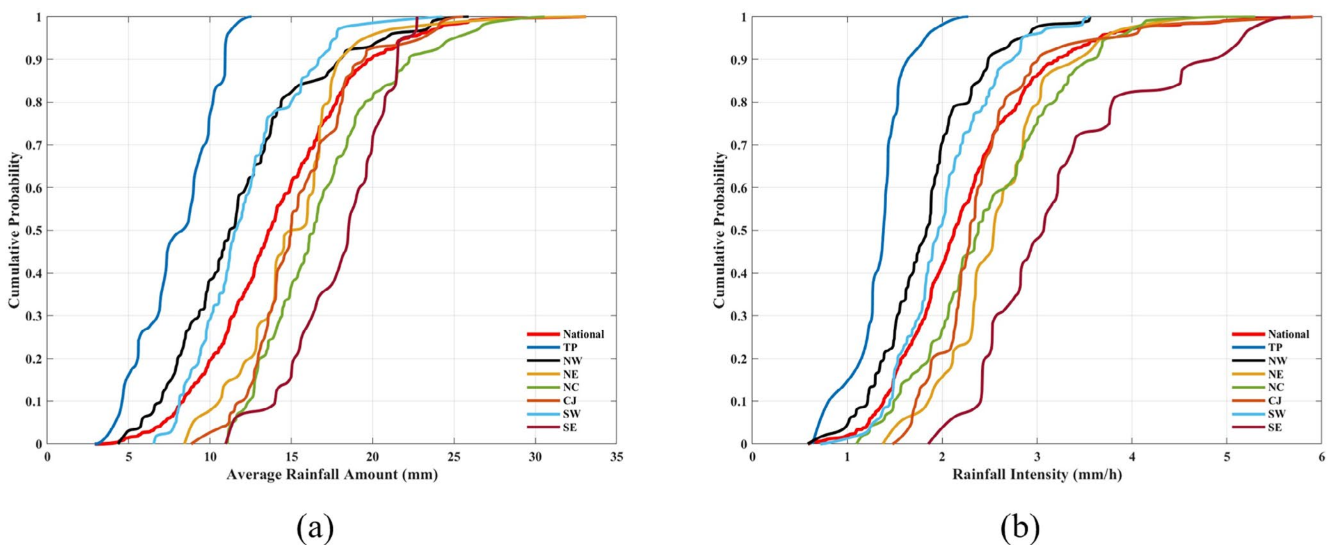

FIGURE A2 | Empirical cumulative distribution function (CDF) of rainfall. (a) Station-averaged rainfall amount, (b) Station-averaged rainfall intensity.

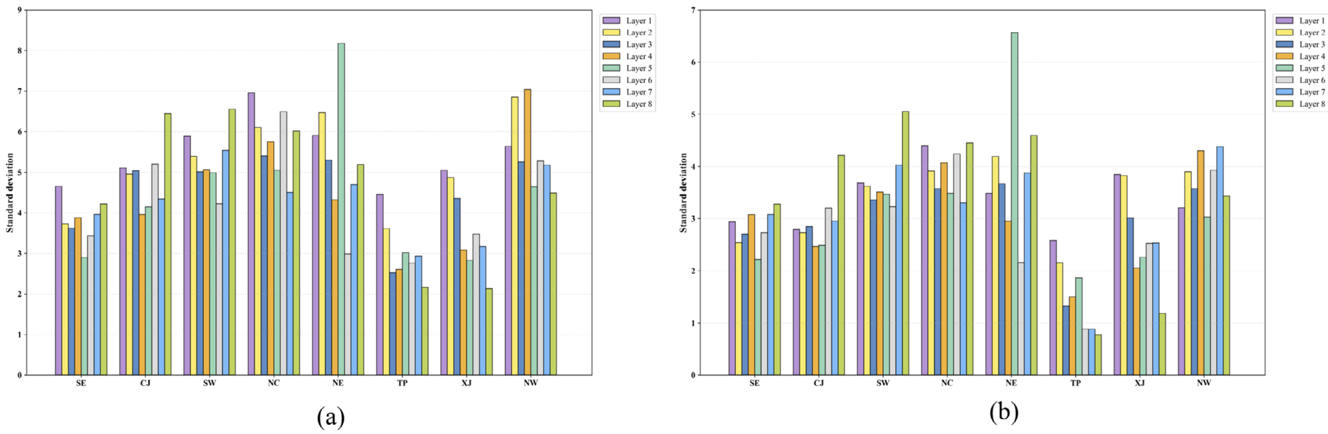

FIGURE A3 | Standard deviations of rainfall-induced SWD indices at the regional scale. (a) Soil water increment (ASWI), (c) Maximum rate of the soil wetting curve (Smax).

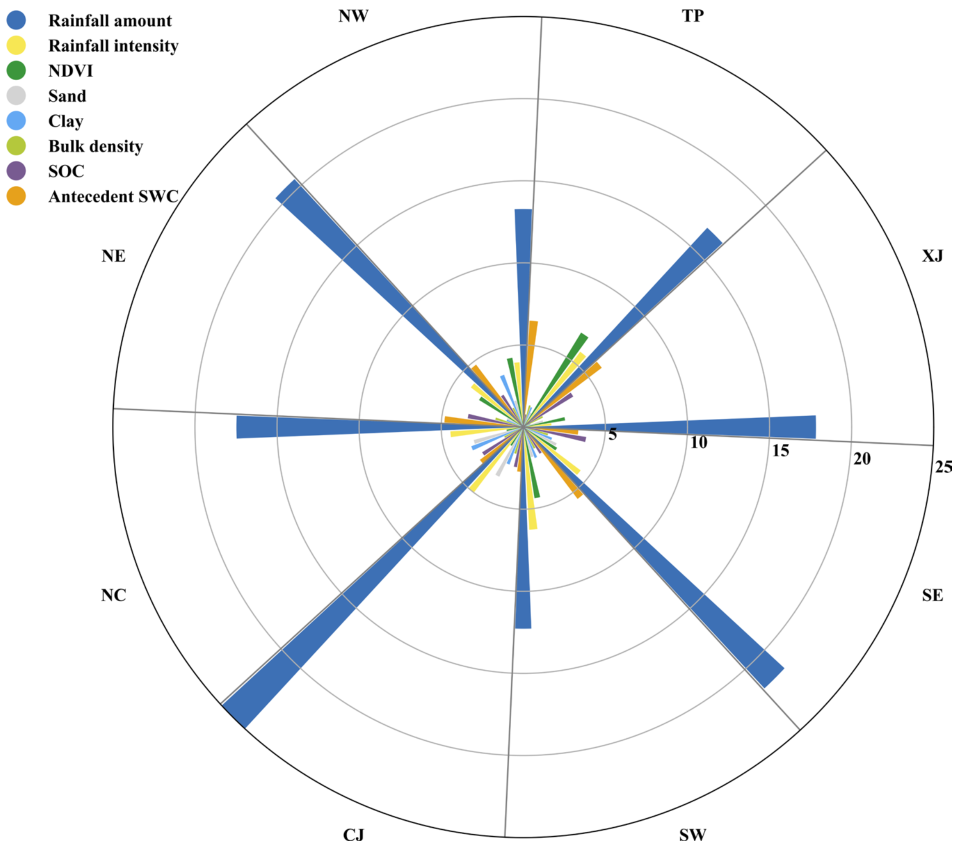

FIGURE A4 | Temporal controls of preferential flow (PF) occurrence in eight subregions.

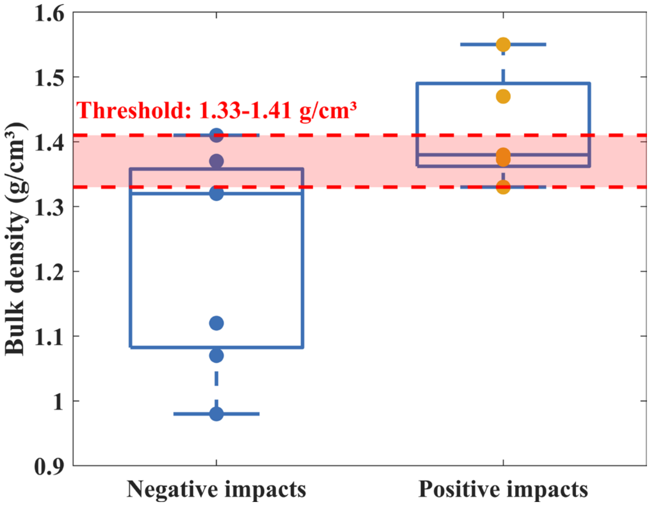

FIGURE A5 | Threshold of bulk density with different impacts on PF frequency.